Lesson Planning

Complete Lesson Plan Book

Class: ICS · Class 12

Subject: Computer Science

Medium: English

Academic Year 2026-27

This book is prepared as a teaching guide for teachers.

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Unit 1: Introduction to Computer Networks: LAN, WAN and the Internet

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Unit 1: Introduction to Computer Networks: LAN, WAN and the Internet

Learning Objectives

• Students will define a computer network and state two benefits of connecting computers.

• Students will distinguish between LAN, MAN and WAN with one example of each.

• Students will explain in simple terms how the Internet connects networks around the world.

Key Concepts

• A computer network is a group of computers connected so they can share data and resources such as files and printers.

• A LAN (Local Area Network) covers a small area like one school building or one home.

• A MAN (Metropolitan Area Network) covers a city, for example a cable network across a town.

• A WAN (Wide Area Network) covers a large area, even whole countries; the Internet is the largest WAN.

• Networks need hardware such as cables, switches and routers, and rules called protocols so computers can understand each other.

Teaching Methodology (Step by Step)

1. Ask students how a school's computers share one printer, and use their answers to define a network.

2. Draw three diagrams on the board: one room (LAN), one city (MAN) and many countries joined together (WAN).

3. Explain the Internet as a network of networks, with undersea cables and routers passing data along.

4. Compare sharing a file by USB stick with sharing it over a network, and list the time saved.

5. End with a quick check: students name the network type for a home Wi-Fi, a city cable network and the Internet.

Classroom Activities

• Network mapping: students draw the network of their school or home and label its devices.

• Type match: cards with examples (home Wi-Fi, city cable TV, the Internet) are matched to LAN, MAN and WAN.

• Speed debate: two groups argue whether sharing files over a network is better than using a USB stick, with reasons.

Assessment Questions

1. Define a computer network and give two benefits of using one.

2. A school's computer lab and a city's cable network: which is a LAN and which is a MAN? Give a reason for each.

3. Why is the Internet called a network of networks?

Homework

• Draw your home network, label the router and at least two connected devices, and write one sentence about what they share.

• Write three differences between a LAN and a WAN.

Time Allocation

40 minutes

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